

NGUYEN HOANG THIEN

Backend Software Engineer | Go & Java | Distributed Systems | Real-Time Messaging & WebRTC

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Professional Summary

Backend-focused Software Engineer with **3+ years of experience** building production services in Go and Java for high-concurrency, real-time, loyalty, CRM/ERP, and SaaS products. Hands-on with RESTful APIs, gRPC, WebSockets, asynchronous processing, event-driven workflows, SQL/NoSQL databases, Redis caching and pub/sub, Kafka, and RabbitMQ. Built reward systems sustaining **5,000+ simultaneous requests**, finance workflows processing **\$100K+ per month**, and data pipelines handling **1M+ rows**. Strong in translating product requirements into reliable backend designs, maintaining data consistency, optimizing performance, tracing production issues, and delivering end-to-end features with React when needed.

Technical Skills

Languages: Go, Java, TypeScript / JavaScript, Python, C
Backend & Architecture: RESTful APIs, gRPC, Spring Boot, WebSockets, concurrency, asynchronous processing, service-oriented design, event-driven architecture
Data & Messaging: PostgreSQL, MongoDB, Redis, Elasticsearch, Kafka, RabbitMQ, SQL/NoSQL, caching, pub/sub
Real-Time & Networking: WebRTC, ion-sfu, signaling, ICE/SDP, HTTP, WebSocket, client-server communication
Frontend: React, Next.js, TypeScript, real-time UI, dashboards, internal tools
DevOps & Engineering: Docker, Kubernetes, Nginx, Linux, CI/CD, Clean Architecture, data consistency, structured logging, performance optimization, Agile / Scrum

Professional Experience

VinSmart Future	May 2026 - Present
Software Engineer - Backend, Loyalty & Digital Business Ho Chi Minh City, Vietnam	
<i>Focus: Java/Go backend services, loyalty, rewards, promotions, business rules, and cross-system workflows</i>	
- Develop and maintain backend services for Loyalty Biz, covering core loyalty, member rewards, and promotion workflows. - Implement REST APIs and business rules in Java and Go, translating product requirements into maintainable service components. - Model workflow states, validations, and edge cases for reward and promotion operations, focusing on data consistency and predictable behavior. - Integrate loyalty features with surrounding internal systems and coordinate cross-team delivery of shared business flows. - Work with product and business stakeholders to clarify rules, acceptance criteria, and technical trade-offs before implementation. - Contribute to initial backend work for the real-estate business domain while remaining primarily responsible for Loyalty Biz.	
Gearment	Dec 2024 - May 2026
Software Engineer - CRM/ERP & Back-Office Platform Ho Chi Minh City, Vietnam	
<i>Focus: Go backend services, order/finance state machines, Kafka integrations, automation, and data consistency</i>	
- Owned the full order lifecycle in Go - creation, hold, modification, cancellation, and refund - for back-office teams managing 10,000+ orders/month. - Engineered G-Credit and tiered-pricing workflows processing \$100K+ in monthly transaction value, enforcing consistent approval, billing, and state transitions. - Translated order and finance requirements from operations and finance teams into APIs, validation rules, state-machine flows, and back-office automation. - Implemented structured event logging and order-state traceability, reducing mean incident-investigation time by ~50%. - Built Kafka-backed partner integrations and webhook flows with retry/replay support, improving external communication reliability by ~30%. - Optimized asynchronous import/export pipelines for datasets exceeding 1M rows, cutting peak memory pressure by ~45% and removing timeout failures. - Automated recurring back-office jobs and standardized ERP module patterns, reducing manual execution and making new workflows easier to extend.	

Software Engineer - Go Backend & React

Focus: high-concurrency Go services, real-time messaging, Redis pub/sub, SaaS modules, and React interfaces

- Built a high-concurrency reward-claiming backend in Go sustaining **5,000+ simultaneous requests** during peak Web3 campaigns and live-stream events with zero data inconsistency or downtime.
- Developed **8+ SaaS management modules** across Go backend services and React interfaces, covering order flows, permissions, and transactional operations using Clean Architecture.
- Delivered real-time chat, video, and user workflows across backend services and React interfaces using WebSockets and Redis pub/sub.
- Implemented event flows for chat, presence, and session synchronization across backend instances, reducing session drop rate by **~35%** and message latency by **~20%**.
- Integrated Lido and DWF Labs protocols for automated reward distribution, reducing manual reconciliation effort by **~60%** through on-chain event handling.
- Applied Clean Architecture and concurrency-safe processing to isolate domain logic and protect transactional consistency under load.

Selected Projects

Real-Time Messaging & Calling Platform - Chat, Voice & Video Communication

2024

Stack: Go, React, TypeScript, ion-sfu, WebRTC, WebSockets, Redis, PostgreSQL, Docker

- Designed a real-time communication platform supporting one-to-one/group chat and multi-party voice/video calls, separating messaging, signaling, presence, and media flows.
- Built WebSocket APIs for messages, conversation history, typing indicators, online presence, delivery/read states, and incoming-call notifications.
- Integrated ion-sfu as the SFU media server for WebRTC calls, managing rooms, participants, media tracks, mute/camera state, and call lifecycle.
- Implemented signaling for invite/accept/reject, room join/leave, SDP and ICE exchange, reconnection, and call-state recovery after interrupted connections.
- Used Redis pub/sub to synchronize messages, presence, call events, and participant state across backend instances.
- Defined event contracts and room/session state transitions to keep messaging, presence, and call workflows consistent during reconnects and cross-instance processing.
- Persisted conversation, message, and room metadata in PostgreSQL while keeping ephemeral presence and call state in Redis.
- Developed React and TypeScript interfaces for conversation threads, incoming-call notifications, participant layouts, and in-call controls across desktop and mobile browsers.

Linkable - AI Data Analyst Assistant - Internal AI / Business Intelligence Workflow Tool

2024

Stack: Openclaw, MCP, isolated execution pods, Playwright, Chrome DevTools, ClickUp, Metabase

- Built a secure internal AI analyst platform enabling **10+ teams** to query business data, review sprint progress, and track workflows without repeated manual support.
- Integrated MCP with Metabase, ClickUp, and internal repositories to combine BI insights, execution tracking, and project context in one AI-driven workflow.
- Engineered isolated execution pods and multi-tenant controls to maintain strict privacy boundaries across teams.
- Developed Playwright and Chrome DevTools pipelines for autonomous URL-based scraping, parsing, and analysis.
- Implemented ClickUp SSO and multi-provider LLM profile management for easier adoption by non-technical stakeholders.

Education

FPT University - B.Sc. Software Engineering

Dec 2020 - Dec 2024

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, Computer Networks, Database Systems

Certifications

- A Practical Guide to Kubernetes - Educative
- MCP Fundamentals for Building AI Agents - Educative
- The Complete Guide to Spring 6 & Spring Boot 3 - Educative
- Claude Code in Action - Anthropic / Skilljar
- User Experience Research and Design